Random Variables

1. The following table provides a probability distribution for the random variable X.

X	3	6	9
$f(X)$	0.25	0.5	0.25

   (a) Compute the expected value of X.

   (b) Compute the variance of X.

   (c) Compute the standard deviation of X.

2. A volunteer ambulance service handles 0 to 5 service calls on any given day. The probability distribution for the number of service calls is as follows.

Number of service calls	0	1	2	3	4	5
probability	0.10	0.15	0.30	0.20	0.15	0.10

   (a) What is the expected number of service calls?

   (b) What us the variance in the number of service calls?

   (c) What is the standard deviation?

3. If the p.d.f of a random variable X is

$$f(x)=\begin{cases} kx & 0\le x\le 5 \\ 0 & \text{elsewhere} \end{cases}$$

   (a) Calculate k

   (b) Calculate $P(1\le X\le 3)$, $P(2\le X\le 4)$, $P(X\le 3)$

4. People visiting video rental store often rent more than one DVD at a time. The probability distribution for DVD rentals per customer at Video To Go is given below:

X	0	1	2	3	4	5
$P(X)$	0.03	0.50	0.24	?	0.07	0.04

There is a limit of 5 videos per customer, so nobody ever rents more than 5 DVDs. The random variable X is $X = \text{number of DVD's a customer rents}$. The values that X can assume are: 0, 1, 2, 3, 4, 5.

(a) Find the probability that a customer rents three DVDs.

(b) Find the probability that a customer rents at least 4 DVDs.

(c) Find the probability that a customer rents at most 2 DVDs.

5. A discrete random variable, X, has a probability distribution function defined as

$$f(X=x)=\begin{cases} c\,x^2 & x=1,2,3,4,5 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the constant c;

(b) Calculate the probability that X will be less than 4 but greater than 1;

(c) Let X_1 and X_2 be two independent random variables from the same distribution as above. Calculate the probability that X_1 is less than 3 and X_2 greater than 2;

(d) Calculate the mean and variance for this distribution.

5. The probability density function of a random variable X is given by

$$f(X=x)=\begin{cases} \frac{3}{14}(2+x+x^2-x^3) & 0<x<2 \\ 0 & \text{elsewhere} \end{cases}$$

(a) Calculate the mean value for this distribution.

(b) Calculate the position of the mode for this distribution.

6. Replicate a selection of questions from 1-5 in **R**. Can you create functions for the distributions and draw plots of them. Can you compute the probabilities? Do your written and computed solutions correspond?